EIGHT EFFECTIVE MATHEMATICS TEACHING PRACTICES

A FRAMEWORK FOR MEANINGFUL MATHEMATICS LEARNING

These eight practices work together to create equitable and meaningful mathematics learning for all students.

1 ESTABLISH MATHEMATICS GOALS TO FOCUS LEARNING

Set clear mathematics goals that guide instructional decisions and focus learning.



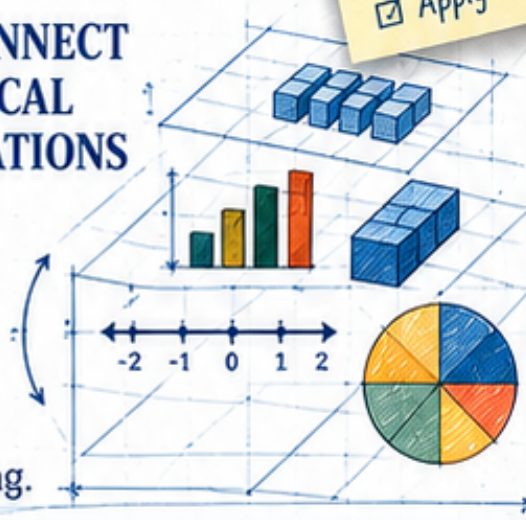
2 IMPLEMENT TASKS THAT PROMOTE REASONING AND PROBLEM SOLVING

Engage students in meaningful and challenging tasks that promote mathematical reasoning and problem solving.



3 USE AND CONNECT MATHEMATICAL REPRESENTATIONS

Help students use and connect multiple representations (diagrams, symbols, contexts, and others) to deepen understanding.



4 FACILITATE MEANINGFUL MATHEMATICAL DISCOURSE

Create opportunities for students to share, compare, and justify their mathematical ideas.



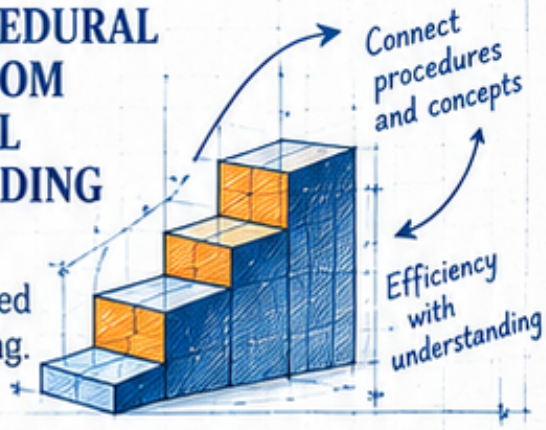
5 POSE PURPOSEFUL QUESTIONS

Ask questions that reveal and advance student thinking.



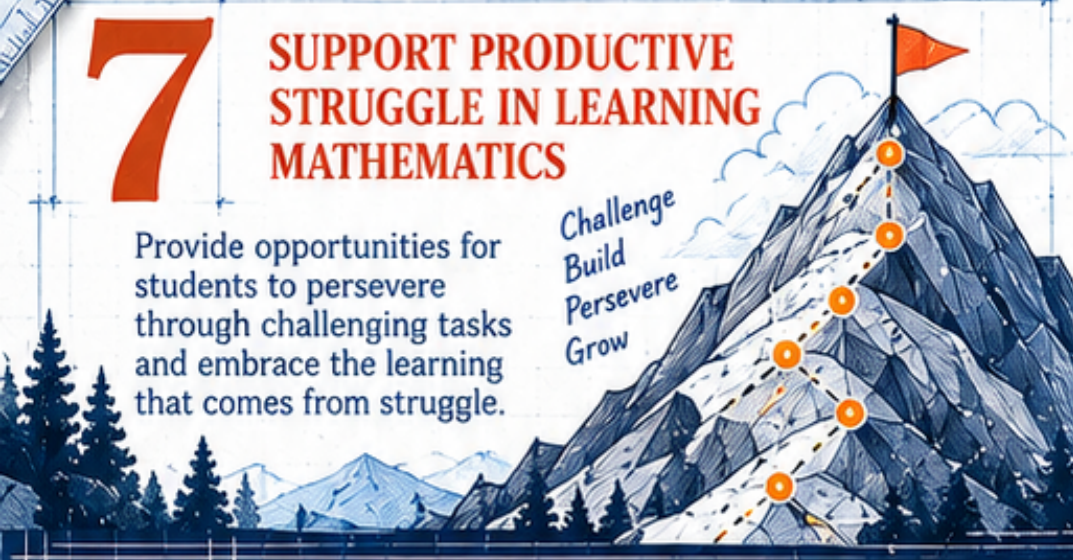
6 BUILD PROCEDURAL FLUENCY FROM CONCEPTUAL UNDERSTANDING

Develop efficient, accurate procedures that are grounded in conceptual understanding.



7 SUPPORT PRODUCTIVE STRUGGLE IN LEARNING MATHEMATICS

Provide opportunities for students to persevere through challenging tasks and embrace the learning that comes from struggle.



8 ELICIT AND USE EVIDENCE OF STUDENT THINKING

Gather and interpret evidence of student thinking to inform instructional decisions.



← ALL EIGHT PRACTICES → WORK TOGETHER ←

to create equitable and meaningful mathematics learning for all students.

Source: National Council of Teachers of Mathematics. (2014). *Principles to Actions: Ensuring Mathematical Success for All*. NCTM.
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